

# LECTURE-20

**Chromosome Structure:** A chromosome is made of  $\text{DNA}$  (Deoxyribonucleic Acid), which is organized into **Base Pairs**. Structurally, it consists of a **Centromere**, a short arm ( $\text{p arm}$ ), and a long arm ( $\text{q arm}$ ), with **Telomeres** at the ends<sup>1111</sup>. A replicated chromosome consists of two sister **Chromatids**<sup>2</sup>.

**Recombination (Crossing Over):** This is the **recombination between 2 homologous chromosomes** (one Paternal and one Maternal)<sup>3</sup>.

- It occurs after  $\text{DNA}$  replication<sup>4</sup>.
- It is the **basis of genetic diversity** amongst individuals of a species<sup>5</sup>.
- The result is a set of **recombined chromosomes**<sup>6</sup>.

Here is a detailed, copy-paste format explanation of the provided slides, covering all figures and concepts related to Mendel's Second Law, extensions of inheritance, and concepts of evolution.

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## Page 1: Law of Segregation (Review)

This page reviews the fundamental results and terminology of a **monohybrid cross**, which led to the Law of Segregation.

- **Phenotype vs. Genotype**<sup>1</sup>:
  - **Phenotype** = organism's **appearance** or **observable traits**<sup>2</sup>.
  - **Genotype** = **genetic makeup** or **allele combinations**<sup>3</sup>.
- **$\text{F}_2$  Ratios (from  $\text{F}_1$  cross  $\text{Pp} \times \text{Pp}$ )**:
  - **Phenotype Ratio: 3 Purple : 1 White** (Ratio 3:1)<sup>4444444444</sup>.
  - **Genotype Ratio: 1  $\text{PP}$  : 2  $\text{Pp}$  : 1  $\text{pp}$**  (Ratio 1:2:1)<sup>5555</sup>.
    - $\text{PP}$  is **homozygous** (Purple)<sup>6</sup>.
    - $\text{Pp}$  is **heterozygous** (Purple)<sup>7777</sup>.

- $\text{pp}$  is **homozygous** (White)<sup>8</sup>.
- **Mendel's Insight:** Mendel inferred these laws without knowing concepts like chromosomes, genes, meiosis, or mitosis<sup>9</sup>.
- **Breeding Consequences**<sup>10101010</sup>:
  - **Homozygous inbreeding** means offspring will have the genotype and phenotype of the parent<sup>11111111</sup>.
  - **Heterozygous breeding** means offspring will have a mix of parents' and grandparents' genotype and phenotype<sup>12</sup>.
- **Test Cross Necessity**<sup>13</sup>:
  - If you only know the phenotype (e.g., purple flowers), the genotype is uncertain ( $\text{PP}$  or  $\text{Pp}$ )<sup>14</sup>.
  - A **Test cross** (mating with a known true-breeding recessive,  $\text{pp}$ ) is needed to determine the unknown genotype by tracking offspring for multiple generations<sup>15</sup>.

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## Page 2, 3, & 15: Introduction to Two Traits, Chromosomes, and Recombination

These pages transition from single-gene inheritance (Law of Segregation) to multiple-gene inheritance (Law of Independent Assortment) and introduce the physical basis for genetic variation.

### Moving to Two Traits

- The Law of Segregation was deduced by following the inheritance patterns of single traits controlled by one gene<sup>16</sup>.
- The next question for Mendel was: **What about 2 traits controlled by 2 genes or more?**<sup>17</sup>.
- **Gene Location and Linkage**<sup>18</sup>: The relationship between the genes depends on their location:
  - The 2 traits can be on **different chromosomes** (e.g., Chromosome  $\text{A}$  and Chromosome  $\text{B}$ )<sup>19</sup>.

- Or, they can be on the **same chromosome** (e.g., Chromosome  $\text{C}$  and Chromosome  $\text{D}$ ).
- If on the same chromosome, they can be **close or far from each other**.

## The Mechanism of Recombination (Crossing Over)

The diagram illustrates the structural basis of chromosomes and the process of crossing over.

- **Chromosome Structure:** A chromosome is made of **DNA (Deoxyribonucleic Acid)**, composed of **Base Pairs**, and structurally features a **Centromere**, a short arm ( $\text{p}$  arm), and a long arm ( $\text{q}$  arm), with **Telomeres** at the ends. A replicated chromosome consists of two sister **Chromatids**.
- **Recombination/Crossing Over:**
  - This is the **recombination between 2 homologous chromosomes** (one Paternal and one Maternal).
  - It occurs after **DNA replication**.
  - It is a physical exchange of genetic material between non-sister chromatids.
  - It **occurs in all species when sperm and eggs are produced** (during meiosis).
  - It is the **basis of genetic diversity** amongst individuals of a species.
  - The result is **recombined chromosomes**, carrying a mix of paternal and maternal alleles.

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## Page 4 & 5: Dihybrid Cross Setup

This section sets up Mendel's dihybrid cross experiment by reviewing the characteristics and assigning gene symbols.

- **Mendel's Monohybrid Results (Table 14.1)**: Mendel knew the dominant and recessive traits from his  $\text{F}_1$  crosses, all of which resulted in an approximate **3:1 phenotypic ratio** in the  $\text{F}_2$ .

[illegible]

- **The Problem:** In nature, multiple traits occur together (e.g., short plant with round seeds, wrinkled and yellow)<sup>35</sup>.
- **The Experiment:** Mendel followed **2 traits at a time**<sup>36</sup>, using traits he already characterized, such as **Seed color** (Yellow vs. Green) and **Seed shape** (Round vs. Wrinkled)<sup>37</sup>.
- **Allele Symbols**<sup>38</sup>:
  - **Seed Color: Yellow** ( $\text{Y}$ , dominant) and **Green** ( $\text{y}$ , recessive)<sup>39</sup>.
  - **Seed Shape: Round** ( $\text{R}$ , dominant) and **Wrinkled** ( $\text{r}$ , recessive)<sup>40</sup>.
- **Diploid Genotype Task**<sup>41</sup>: The task is to assign diploid genotypes to four phenotypes (possible answers were requested to be written in the spaces, but based on the diagram: Yellow-Round:  $\text{YYRR}$  or  $\text{YyRr}$ , Green-Round:  $\text{yyRR}$  or  $\text{yyRr}$ , Yellow-Wrinkled:  $\text{YYrr}$  or  $\text{Yyrr}$ , Green-Wrinkled:  $\text{yyrr}$ ).

## Page 6: Hypothesis of Dependent Assortment (The Wrong Prediction)

This page outlines the *incorrect* hypothesis—the idea that traits must be inherited together—and shows why the actual results refute it.

- P Generation Cross:** Mendel crossed two **true-breeding** plants: **Yellow-Round seeds** ( $\text{YYRR}$ )  $\times$  **Green-Wrinkled seeds** ( $\text{yyrr}$ ).
  - Each  $\text{P}$  plant is diploid and has two copies of each allele (e.g.,  $\text{YY}$  and  $\text{RR}$ ).
  - Each  $\text{P}$  plant produces **one type of haploid gamete**:  $\text{YR}$  or  $\text{yr}$ .
- $\text{F}_1$  Generation:** Fertilization produces only one diploid combination:  $\text{YyRr}$ .

- Since  $\text{Y}$  and  $\text{R}$  are dominant, the  $\text{F}_1$  plant has **Yellow-Round seeds**.
- **Hypothesis of Dependent Assortment**:
  - This hypothesis states that **dominant or recessive traits always get inherited together**.
  - Prediction: The  $\text{F}_1$  plant ( $\text{YyRr}$ ) would only produce two types of gametes:  $\text{YR}$  and  $\text{yr}$ .
- **$\text{F}_2$  Prediction (Based on Dependent Assortment)**:
  - **$\text{F}_2$  Genotypes**:  $\text{YYRR}$ ,  $\text{YyRr}$ , or  $\text{yyrr}$ .
  - **Phenotypic Ratio: 3:1** (3 Yellow-Round : 1 Green-Wrinkled).
- **Conclusion: THIS IS NOT THE RESULT MENDEL OBTAINED**.

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## Page 7 & 8: Hypothesis of Independent Assortment (The Correct Prediction)

These pages illustrate and explain the correct hypothesis—the Law of Independent Assortment—which matched Mendel's experimental results.

- **Hypothesis of Independent Assortment**: This law states that **two or more genes assort independently**—that is, each pair of alleles segregates independently of each other pair of alleles during gamete formation.
- **$\text{F}_1$  Gametes (Independent Assortment)**: If traits are inherited separately, the  $\text{F}_1$  plant ( $\text{YyRr}$ ) will produce **four types of gametes** in equal proportion ( $\frac{1}{4}$  each):
  - $\frac{1}{4}$   $\text{YR}$
  - $\frac{1}{4}$   $\text{Yr}$
  - $\frac{1}{4}$   $\text{yR}$
  - $\frac{1}{4}$   $\text{yr}$
- **$\text{F}_2$  Results (Punnett Square)**: Combining these four gamete types in a Punnett Square results in 16 possible  $\text{F}_2$  offspring combinations.

- **$\text{F}_2$  Phenotypic Ratio**<sup>59</sup>: The resulting phenotypic ratio is **9:3:3:1**<sup>60</sup>:
  - $\frac{9}{16}$  **Yellow-Round** (Parental)<sup>61</sup>
  - $\frac{3}{16}$  **Yellow-Wrinkled** (New combination)<sup>62</sup>
  - $\frac{3}{16}$  **Green-Round** (New combination)<sup>63</sup>
  - $\frac{1}{16}$  **Green-Wrinkled** (Parental)<sup>64</sup>
- **Key Discovery**<sup>65</sup>: The appearance of **two new traits in  $\text{F}_2$**  (**Green-Round and Yellow-Wrinkled**)<sup>66</sup> proved that the genes for color ( $\text{Y}/\text{y}$ ) and shape ( $\text{R}/\text{r}$ ) had separated from each other and combined independently<sup>67</sup>. **THIS IS THE RESULT MENDEL OBTAINED**<sup>68</sup>.

## Page 9: Law of Independent Assortment (Contextualization)

This page clarifies the conditions under which the Law of Independent Assortment applies, linking it to chromosome location.

- **Assortment Condition**<sup>69</sup>: The law applies only to genes (allele pairs) located on **non-homologous chromosomes**<sup>70</sup>.
- **Linkage Exception**<sup>71</sup>: Alternatively, it applies to genes that are **very far apart on the same chromosome** (because recombination is highly likely between them)<sup>72</sup>.
- **Chromosomal Locations**<sup>73</sup>: The independent assortment principle is illustrated by genes on different chromosomes (A and B) or on the same chromosome but far apart (C and D)<sup>74</sup>.

## Page 10, 11, & 12: Extensions of Mendelian Genetics: Human Blood Groups

These pages use the example of human blood groups to illustrate non-Mendelian inheritance patterns like multiple alleles and codominance, and to connect genes to chromosomes.

## Red Blood Cell (RBC) Antigens (Page 10)

- The surface of RBCs has many antigens, including carbohydrates, glycoproteins, and proteins<sup>75</sup>.
- \$\text{ABO}\$ Antigens:** The "\$\text{A}\$" or "\$\text{B}\$" antigen is a **carbohydrate**<sup>76</sup>.  
The oligosaccharide sequence determines if the antigen is \$\text{A}\$ or \$\text{B}\$<sup>77</sup>.
- \$\text{Rh}\$ Antigen:** The "\$\text{Rh}\$" antigen is a **protein** (with 49 \$\text{Rh}\$ variations)<sup>78</sup>.
- Membrane Structure :** Antigens belong to different protein types, such as **Single-pass proteins** (\$\text{ABO}\$, \$\text{Hh}\$, \$\text{MNS}\$, \$\text{Kell}\$) or **Multi-pass proteins** (\$\text{Kidd}\$, \$\text{Duffy}\$) or **GPI-linked proteins**<sup>79</sup>.

## The $\text{ABO}$ Gene (Page 11)

- The **A** gene indirectly encodes the A blood group antigens (carbohydrates)<sup>80</sup>.
- **Multiple Alleles**<sup>81</sup>: The ABO locus has **three main allelic forms**: A, B, and O<sup>82</sup>.
- **Dominance and Molecular Basis**<sup>83</sup>:
  - A allele is **dominant**<sup>84</sup>.
  - B allele is **also dominant**<sup>85</sup>.
  - O allele is **recessive**<sup>86</sup>.
  - A and B alleles encode a **glycosyltransferase** enzyme that converts an antigen precursor (H) into the A and B antigen<sup>87</sup>.
  - The A and B alleles result in enzymes that differ by only **four amino acids** due to a few changes in the DNA sequence<sup>88</sup>.

- The **O** allele has a DNA sequence change that introduces a **stop codon**, resulting in an **inactive enzyme** that cannot convert the H precursor into A or B antigen.

**Two-Gene Inheritance: ABO and Rh (Page 12)**

- The ABO gene is on **Chromosome 9**.
- The Rh gene is on **Chromosome 1**.
- **Rh Dominance:** Rh<sup>+</sup> allele is **dominant** and Rh<sup>-</sup> allele is **recessive**.
- **Possible Genotypes:** Since ABO and Rh genes are on **different, non-homologous chromosomes**, they assort independently. The list of possible diploid genotype combinations (e.g., AARh<sup>+</sup>Rh<sup>+</sup>, AORh<sup>-</sup>Rh<sup>-</sup>) shows the combinations possible for two independently inherited traits.
- **Analogy to Mendel's Cross**: The genotype most similar to Mendel's P generation cross (YYRR  $\times$  yyrr) would be one involving homozygotes for the dominant and recessive alleles of both genes, such as AARh<sup>+</sup>Rh<sup>+</sup> (equivalent to YYRR) crossed with OORh<sup>-</sup>Rh<sup>-</sup> (equivalent to yyrr).

**Page 13: Summary of Inheritance Patterns (Allelic Relationships)**

This table summarizes five different ways that alleles of a single gene can interact to determine a phenotype.

| Relationship among alleles of a single gene | Description | Example |
|---------------------------------------------|-------------|---------|
|                                             |             |         |



|                                              |                                                                                                     |                                                                        |                                                                                                                                                                                              |                                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| <b>Complete dominance</b> of one allele      | Heterozygous phenotype is the same as that of the homozygous dominant <sup>97</sup> .               |                                                                        | $\text{Pp}$ is the same phenotype as $\text{PP}$ <sup>98</sup> .                                                                                                                             |                                                                                                   |
| <b>Incomplete dominance</b> of either allele | Heterozygous phenotype is <b>intermediate</b> between the two homozygous phenotypes <sup>99</sup> . |                                                                        | $\text{C}^{\text{R}}\text{C}^{\text{W}}$ (pink) is intermediate between $\text{C}^{\text{R}}\text{C}^{\text{R}}$ (red) and $\text{C}^{\text{W}}\text{C}^{\text{W}}$ (white) <sup>100</sup> . |                                                                                                   |
| <b>Codominance</b>                           |                                                                                                     | <b>Both phenotypes are expressed</b> in heterozygotes <sup>101</sup> . |                                                                                                                                                                                              | $\text{I}^{\text{A}}\text{I}^{\text{B}}$ (Type AB blood, both A and B expressed) <sup>102</sup> . |
| <b>Multiple alleles</b>                      | In the population, some genes have <b>more than two alleles</b> <sup>103</sup> .                    |                                                                        | $\text{ABO}$ blood group alleles $\text{I}^{\text{A}}$ , $\text{I}^{\text{B}}$ , $\text{i}$ <sup>104</sup> .                                                                                 |                                                                                                   |

|            |  |                                                                |  |                                    |
|------------|--|----------------------------------------------------------------|--|------------------------------------|
| Pleiotropy |  | One gene affects multiple phenotypic characters <sup>105</sup> |  | Sickle-cell disease <sup>106</sup> |
|------------|--|----------------------------------------------------------------|--|------------------------------------|

### Page 14: Polygenic Inheritance (Three Genes)

This page extends the concept of independent assortment to **three genes**, illustrating how multiple genes can influence a single characteristic, such as skin pigmentation.

- Three-Gene Cross:** The cross is  $\text{AaBbCc} \times \text{AaBbCc}$ <sup>107</sup>
- Gamete Production:** Since three genes are involved, there are  $2^3 = 8$  possible gamete combinations, each with a  $\frac{1}{8}$  probability<sup>108</sup>.
- The Punnett Square:** Combining 8 types of sperm and 8 types of eggs results in a  $8 \times 8 = 64$  square Punnett table<sup>109</sup>.
- Phenotype Distribution:** This cross results in a **wide range of phenotypes** (e.g., skin color, which is controlled by multiple genes, demonstrating **polygenic inheritance**).
- Phenotype Ratio:** The phenotypes follow a distribution based on the total **Number of dark-skin alleles** (from 0 to 6)<sup>110</sup>:
  - 0 alleles:  $\frac{1}{64}$
  - 1 allele:  $\frac{6}{64}$
  - 2 alleles:  $\frac{15}{64}$
  - 3 alleles:  $\frac{20}{64}$
  - 4 alleles:  $\frac{15}{64}$
  - 5 alleles:  $\frac{6}{64}$
  - 6 alleles:  $\frac{1}{64}$
  - This pattern forms a classic bell curve distribution, typical of traits controlled by multiple independent genes.

### Page 16, 17, & 18: Genetic Linkage and Mapping

These pages discuss the concept of **Genetic Linkage**, which is an exception to the Law of Independent Assortment.

## Genetic Linkage

- **Genes on Different Chromosomes:** Genes on different chromatids are **not physically contiguous** on the same DNA molecule and can be **inherited independently** (Mendel's Law applies)<sup>111</sup>.
- **Genes on the Same Chromosome**<sup>112</sup>:
  - Genes on the same chromatid are **physically contiguous**<sup>113</sup>.
  - They can be inherited independently if they are **far from each other** (due to frequent crossing over)<sup>114</sup>.
  - They are likely to be inherited independently **extremely rarely** if they are **near to each other**<sup>115</sup>.
  - The **closer they are to each other, the less likely it is for them to be inherited independently** (they are **linked**)<sup>116116116116</sup>.

## Recombination Frequency and Gene Mapping

- **Question:** If a cross produced 100 progeny, in how many of these progeny do two traits manifest together?<sup>117</sup>
- **Answer:** It depends on how close 2 genes are on the same chromosome<sup>118</sup>.
- **Estimating Distances:** The distance between genes on chromosomes is estimated using **recombination frequencies**<sup>119119119119</sup>.
  - If traits are separated (recombined) in **1 out of 100 progeny (1%)**, then the genes controlling those traits are **1 centimorgan ( $\text{cM}$ )** apart<sup>120</sup>.
- **Example Chromosome Map** : The map shows three genes,  $b$ ,  $cn$ , and  $vg$ , on a chromosome<sup>121</sup>:
  - $b$  and  $cn$  have a recombination frequency of **9%**<sup>122</sup>.
  - $cn$  and  $vg$  have a recombination frequency of **9.5%**<sup>123</sup>.
  - The total distance between  $b$  and  $vg$  is  $\approx 17\%$  ( $9\% + 9.5\% = 18.5\%$ , but the measured distance is 17%, indicating complexity not covered in the linear model)<sup>124</sup>.

These final pages introduce the concepts of convergent/divergent evolution and a mechanism for the formation of new species.

- **Convergent Evolution:** Occurs when the **same traits evolve independently multiple times** in species with **different ancestry**.<sup>125</sup>
  - *Example:* *Cereus* sp. (a cactus) and *Euphorbia canariensis* (a euphorb, both having similar fleshy, spined stems)<sup>126</sup>.
- **Divergent Evolution:** Occurs when species with a **common ancestor** evolve different traits over time.
  - *Example 1:* All dog species evolved from a common wolf ancestor.<sup>127</sup>
  - *Example 2:* Elephants and Mammoths evolved from a common ancestor.<sup>128</sup>
  - *Example 3 (Lizards) :* The **Ancestral Lizard (with limbs)** diverged into groups. **Snakes** and the **Eastern glass lizard** evolved "**No limbs**" independently at different points in their lineages (a case of parallel/divergent evolution from an ancestor with limbs)<sup>129</sup>

- **Reproductive Barrier:** This is what **defines a species**<sup>130</sup>.
- **Experiment (Fruit Flies):** An initial population of fruit flies (*Drosophila pseudoobscura*) was divided<sup>131</sup>.
  - Some flies were raised on a **maltose medium**<sup>132</sup>.
  - Others were raised on a **starch medium**<sup>133</sup>.
- **Results (After 40 generations)**<sup>134</sup>:
  - **Mating experiments** showed that flies grown only on starch or maltose prefer to mate with **like-partners** (flies from the same medium)<sup>135</sup>.

- **Table Data (Experimental Group)**<sup>136</sup>: **Starch-Male/Starch-Female** matings (22) and **Maltose-Male/Maltose-Female** matings (20) were much higher than cross-medium matings (9 and 8).
- **Conclusion**: The two groups of flies can, over time, evolve into **two different species**<sup>137</sup> due to the development of a reproductive barrier.
- **Mechanism of Speciation**<sup>138</sup>:
  - **Different Selection Pressures**: Groups experience different selection pressures (in this case, food source), which may change their physiology, behavior, or genes, leading to **reproductive isolation**<sup>139</sup>.
  - **Geographic Separation**: Groups may also be separated **geographically**, which can lead to reproductive isolation<sup>140</sup>.